
Superradiance, spontaneous pair creation, and momentum-space compactness in real-time lattice simulations of relativistic-quantum plasmas

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Abstract

Real-time lattice simulations of scalar quantum electrodynamics (QED) in the classical-statistical regime are a first-principles tool for plasmas in which collective scales overlap relativistic-quantum scales. We delimit the validity domain of the variational, gauge-symmetric lattice scheme by confronting an independent implementation—verified against the original linear-wave and laser-plasma demonstrations—with four strong-field benchmarks that admit sharp analytic predictions. Two benchmarks extend the verified scope. A two-species pair-ion plasma with fully dynamical heavy species reproduces all four branches of the cold relativistic-quantum longitudinal spectrum—including a gapless branch of purely quantum origin whose long-wavelength dispersion is the recoil law $\omega = k^2/(2\sqrt{m_e m_i})$ —to within $0.05 m$ of the tree-level dispersion relations. A wave packet incident on a supercritical electrostatic barrier undergoes superradiant Klein scattering: the reflected charge is amplified to $R = 1.8797$, matching the mode-matching prediction to 0.002%, with the excess negative charge exactly balanced by transmitted antiparticles and total charge conserved to machine precision. Two further benchmarks locate structural boundaries and motivate extensions that remove them. First, spontaneous Schwinger pair creation from a field-free region is absent: measured yields obey $n_k = \varepsilon |\beta_k|^2$, scaling linearly with the seeded mode occupation $\varepsilon/2$ with no ε -independent floor, so the scheme as initialized in prior work produces identically zero pairs at every resolution. Restoring the quantum half, $\varepsilon = 1$, through symplectically normalized vacuum seeding with vacuum-subtracted projections reproduces the exact Sauter-pulse Bogoliubov spectrum mode by mode, including the Schwinger exponent across a factor of two in field strength. Second, under a sustained accelerating field the compact Wilson-line coupling drags every matter mode across the Brillouin zone: the current reverses at $t_{\text{rev}} = \pi/(qE \Delta x)$ at every resolution studied—a Bloch oscillation with no continuum counterpart—so uniform-in-time convergence fails for secular acceleration. A holonomy-resolved extension that couples the spatial-mean gauge potential noncompactly through a spectral matter sector removes the horizon and tracks relativistic saturation to Lorentz factors near 200 at 0.24% accuracy. Both extensions are variational and retain exact charge conservation. Together the four benchmarks show that classical-statistical lattice QED captures stimulated strong-field processes of occupied modes, and provide tested extensions for spontaneous production and secular acceleration.

1 Introduction

When plasmas become dense enough, or electromagnetic fields strong enough, that collective plasma scales are no longer separated from the Compton scale, the standard kinetic description of plasma

*Code, run logs, and derivation scripts reproducing every number in this paper are archived with the submission.

physics—a geometrical-optics reduction of the underlying relativistic-quantum dynamics [Ruiz and Dodin, 2015]—breaks down. Laser-driven electron-positron pair production [Liang et al., 1998, Gahn et al., 2000, Liang et al., 2015, Sarri et al., 2015], wave propagation at frequencies comparable to the electron rest mass, and plasmas in magnetar-strength fields all live in this regime. The semiclassical remedy of inserting prefabricated source terms into kinetic or fluid equations [Berezhiani et al., 1992, Kluger et al., 1998, Schmidt et al., 1998, Roberts et al., 2002, Hebenstreit et al., 2010, 2008, 2009, Duclous et al., 2011, Nerush et al., 2011, Ridgers et al., 2012] cannot describe interference between coexisting processes [Schützhold et al., 2008], and it strains energy–momentum book-keeping precisely when many pairs are produced. A description at the level of interacting quantum fields is then required, and real-time lattice techniques adapted from strong-interaction physics [Wilson, 1974, Creutz, 1980, Yagi et al., 2005, Schwinger, 1961, Keldysh, 1965] offer a practical route: in the classical-statistical regime of high occupation numbers and weak coupling, tree-level dynamics dominates loop corrections [Aarts and Berges, 2002, Mueller and Son, 2004, Jeon, 2005, Berges and Gasenzer, 2007, Berges et al., 2014], and the quantum evolution is faithfully represented by classical field equations advanced over an ensemble of suitably distributed initial conditions [Aarts and Smit, 1999, Gelis and Tanji, 2013, Polkovnikov, 2003, Borsányi and Hindmarsh, 2009].

For plasma applications, Shi et al. [2018] introduced a variational discretization of scalar QED—charged scalar matter on lattice vertices, the gauge potential on edges, Wilson-line covariant differences, and a Yee-type field action—whose discrete $U(1)$ symmetry enforces local charge conservation to machine precision and whose geometric construction satisfies the Bianchi identities exactly. The scheme was demonstrated on linear plasma-wave spectra and on an intense laser interacting with a dense plasma slab, where wakefield acceleration transitions naturally into Schwinger pair production; related structure-preserving formulations have since been extended to spinor QED and to large heterogeneous machines [Chen et al., 2021, 2022], to quantum processors [Sundar et al., 2025, Shi et al., 2024], and the underlying discretization has been analyzed in the numerical-analysis literature [Christiansen and Halvorsen, 2011, Christiansen et al., 2021]. These developments raise a question that the demonstrations themselves do not answer: *which* strong-field phenomena does the classical-statistical lattice representation actually capture, and where—not for reasons of resolution or implementation, but structurally—does it stop?

This paper answers that question quantitatively for four strong-field benchmarks, each chosen to probe a physically distinct regime with a sharp, independently computable prediction, using a single independently developed implementation of the variational scheme that first reproduces both of the original demonstrations. The four benchmarks split naturally into two that *extend the verified scope* of the method and two that *locate its structural boundaries and test extensions that remove them*:

(i) *Multi-species collective modes.* All previous validations of the scheme used a single dynamical species against an immobile neutralizing background. We simulate a two-species pair-ion plasma in which both light and heavy species are dynamical, and recover the complete four-branch cold longitudinal spectrum—a Langmuir-like branch, two pair branches at twice each rest mass, and a gapless branch that has no cold classical counterpart: its quadratic long-wavelength dispersion $\omega \rightarrow k^2/(2\sqrt{m_e m_i})$, which we derive from the quantum-recoil terms of the relativistic-quantum susceptibility, is set by the particle masses alone, with no trace of the plasma frequencies.

(ii) *Superradiant Klein scattering.* Scattering of a charged wave packet off a supercritical electrostatic barrier is the oldest relativistic-quantum scattering problem [Klein, 1929], and its resolution—amplified reflection fed by stimulated pair creation at the barrier—is a stringent test of charge book-keeping in any numerical relativistic-quantum method. The lattice scheme reproduces the mode-matching reflection coefficient $R > 1$ to 0.002% while conserving global charge to better than 10^{-13} , and an exact discrete gauge transformation mapping the static barrier onto a time-ramped vector potential leaves every observable unchanged at machine precision.

(iii) *Spontaneous versus stimulated pair creation.* The laser-plasma demonstration of Shi et al. [2018] produced copious pairs, but—as we show—those pairs are *stimulated*: they grow from modes that are already macroscopically occupied by the plasma. Applying a supercritical Sauter pulse to a field-free region, the classical-statistical scheme yields exactly zero pairs, at every lattice resolution, because $\phi = 0$ is a fixed point of the classical flow; more sharply, seeding the ensemble at a mean occupation $\varepsilon/2$ per mode produces the spectrum $\varepsilon |\beta_k|^2$, linear in ε with no floor, where $|\beta_k|^2$ is the exact Bogoliubov pair spectrum. Nothing in the classical theory selects ε ; the quantum commutator does, and fixing $\varepsilon = 1$ (the Wigner half-quantum) with symplectically normalized lattice mode

functions and vacuum-subtracted projections turns the scheme into a quantitative instrument for spontaneous production, reproducing exact Sauter-pulse spectra mode by mode and the Schwinger exponent over a factor of two in field strength.

(iv) *Secular acceleration and momentum-space compactness.* The Wilson-line coupling makes the lattice theory periodic in $qA\Delta$, a property advertised as a resolution constraint $qA\Delta < 2\pi$ [Shi et al., 2018]. We show that for sustained acceleration this constraint is not merely a step-size rule but a *horizon*: under a constant force qE every matter mode is dragged across the compact Brillouin zone and the current *reverses* at $t_{\text{rev}} = \pi/(qE \Delta x)$ —a Bloch oscillation, familiar from electrons in crystals [Bloch, 1929, Zener, 1934] and cold atoms in optical lattices [Ben Dahan et al., 1996], but here a pure discretization artifact with no continuum counterpart. Refining the lattice postpones the reversal only linearly, so no fixed resolution converges uniformly in time, and the resolution requirement is set by the highest momentum *reached* rather than by any scale of the initial data. We construct a holonomy-resolved extension—the spatial-mean (holonomy) part of the gauge potential couples noncompactly to a spectral matter sector—that is variational, retains exact charge conservation and exact covariance under large gauge transformations, and removes the horizon entirely: it tracks the continuum work-energy flow to Lorentz factors $\gamma \approx 200$, converting a structural failure into an ordinary time-resolution requirement.

The two boundaries are complementary in an instructive way. The first is a boundary of the *ensemble*: classical-statistical evolution transports occupations but cannot create them from nothing, so spontaneous quantum processes require the vacuum’s half-quantum to be present in the initial data. The second is a boundary of the *representation*: a local, gauge-invariant lattice necessarily compactifies kinetic momentum, so secular acceleration eventually exhausts the representable phase space. Both boundaries are invisible to conservation-law diagnostics—charge, Gauss constraint, and gauge invariance remain machine-exact while the physics silently fails—which we regard as the central methodological lesson of this study: structure preservation certifies the discrete dynamics, not the fidelity of the discrete theory to the continuum one.

The paper is organized as follows. Section 2 summarizes the variational scheme, our implementation-verification protocol, the analytic predictions and tolerances for all benchmarks, the failure criteria for the two boundary studies, and the two extensions at the level of the discrete action. Section 3 specifies all numerical parameters and the provenance of every number. Section 4 presents the reproductions and the four benchmark results, and Section 5 synthesizes what they imply about the framework; Section 6 concludes.

2 Methodology

2.1 Variational lattice scalar QED

We work in natural units $\hbar = c = \epsilon_0 = 1$ with metric signature $(+, -, -, -)$. The continuum model is scalar QED for a set of charged Klein–Gordon fields ϕ_ζ (charge coupling q_ζ , mass m_ζ) minimally coupled to a U(1) gauge field,

$$\mathcal{L} = \sum_{\zeta} \left[(\overline{D_\mu \phi_\zeta})(D^\mu \phi_\zeta) - m_\zeta^2 \bar{\phi}_\zeta \phi_\zeta \right] - \frac{1}{4} F_{\mu\nu} F^{\mu\nu}, \quad D_\mu = \partial_\mu - iq_\zeta A_\mu, \quad (1)$$

whose Euler–Lagrange equations are the Klein–Gordon–Maxwell (KGM) system (the scalar model sidesteps lattice fermion doubling [Chandrasekharan and Wiese, 2004] while retaining every plasma-relevant coupling [Eliasson and Shukla, 2011, Shi et al., 2016]) with current $j^\mu = \sum_{\zeta} (q_\zeta/i) [\bar{\phi}_\zeta D^\mu \phi_\zeta - \text{c.c.}]$. In the classical-statistical regime the quantum dynamics is represented by evolving this classical system over an ensemble of initial conditions [Aarts and Berges, 2002, Berges et al., 2014, Gelis and Tanji, 2013].

Following Shi et al. [2018], the fields are discretized on a rectangular spacetime lattice respecting discrete exterior calculus: ϕ lives on vertices, A on edges (stored as the covariant components, e.g. $a_{1,j+1/2}^n = A_1 = -A^x$), and $F = dA$ on faces. Covariant differences use Wilson lines,

$$(D_t \phi)_j^{n+1/2} = \frac{\phi_j^{n+1} e^{-iq \Delta t a_{0,j}^{n+1/2}} - \phi_j^n}{\Delta t}, \quad (D_x \phi)_{j+1/2}^n = \frac{\phi_{j+1}^n e^{-iq \Delta x a_{1,j+1/2}^n} - \phi_j^n}{\Delta x}, \quad (2)$$

and the discrete action per unit cell is the Yee-type form (after Yee [1966]; the same structure-preserving philosophy underlies geometric particle-in-cell methods [Squire et al., 2012, Xiao et al., 2013, 2015])

$$L_d = \sum_{\zeta} \left[|D_t \phi|^2 - |D_x \phi|^2 - |D_y \phi|^2 - |D_z \phi|^2 - m_{\zeta}^2 |\phi|^2 \right] + \frac{1}{2} (E^2 - B^2) - \rho_{\text{bg}} a_0, \quad (3)$$

where E and B are the face-centered field strengths built from first differences of the edge potentials, and the last term couples a static background (jellium) charge density ρ_{bg} where a neutralizing species is idealized as immobile. We simulate one spatial dimension as a 4D lattice with a single periodic transverse cell ($\Delta y = \Delta z = \Delta x$), so transverse link phases enter through $2[1 - \cos(q\Delta y a_2)]/\Delta y^2$ and fields depend on (t, x) only. Extremizing $S_d = \sum_{\text{cells}} \Delta t \Delta x L_d$ yields the discrete Klein–Gordon equation, the discrete Gauss constraint $(e_{1,j+1/2} - e_{1,j-1/2})/\Delta x = \rho_j + \rho_{\text{bg},j}$ with $\rho_j^{n+1/2} = -\sum_{\zeta} (2q_{\zeta}/\Delta t) \text{Im}[\bar{\phi}_j^{n+1} e^{iq_{\zeta} \Delta t a_{0,j}} \phi_j^n]$, and the discrete Maxwell–Ampère law $e^{n+1/2} = e^{n-1/2} + \Delta t (\text{curl } B - j)$ with the point-split currents conjugate to the links $[j_{j+1/2}^x = \sum_{\zeta} (2q_{\zeta}/\Delta x) \text{Im}(\bar{\phi}_{j+1}^n e^{iq_{\zeta} \Delta x a_{1,j+1/2}} \phi_j^n)]$. The discrete U(1) symmetry $\phi \rightarrow \phi e^{iq\alpha}$, $a_{\mu} \rightarrow a_{\mu} + \Delta_{\mu} \alpha$ implies exact local charge conservation; the Faraday law and $\nabla \cdot \mathbf{B} = 0$ hold identically by construction; and the cross-time discrete energy of Shi et al. [2018] is conserved exactly at $q \rightarrow 0$ and to $\mathcal{O}(q\Delta^2)$ otherwise. Time stepping uses the temporal gauge ($a_0^{\text{dyn}} = 0$) throughout; prescribed external backgrounds—a static electrostatic potential $a_0^{\text{ext}}(x)$, a spatially uniform $a_1^{\text{ext}}(t)$, or a pure-gauge ramp—enter the matter Wilson lines only, i.e. the action is the background-field form $S_d[\phi, A^{\text{dyn}}; A^{\text{ext}}]$. Under this convention the e^{-imt} branch of a species with coupling q carries physical charge $-q$; "electrons" are modeled with $q = +0.3$ as in Shi et al. [2018], giving the physical fine-structure constant $q^2/4\pi \approx 1/137$.

2.2 Implementation verification

Because every conclusion of this study rests on the discrete dynamics being the *variational* one, we verify the implementation directly rather than by inspection. An independent routine evaluates S_d as a function of a complete field history; production trajectories are then tested for stationarity by central finite differences with respect to every interior degree of freedom. On randomized two-species field configurations with all external backgrounds active, the largest relative gradient is 1.4×10^{-10} (the finite-difference floor), including the Gauss component $\delta S_d/\delta a_0$; total charge is conserved to 7.4×10^{-15} (relative), the Gauss residual is preserved to 1.5×10^{-13} , gauge-invariant observables after a random discrete gauge transformation of the initial data agree to 7.3×10^{-14} , a free lattice mode propagates at the exact dispersion $\frac{4}{\Delta t^2} \sin^2 \frac{\omega \Delta t}{2} = \frac{4}{\Delta x^2} \sin^2 \frac{k \Delta x}{2} + m^2$ to machine precision, and the discrete energy of an uncharged field is constant to 2.3×10^{-15} , confirming the exact $q \rightarrow 0$ conservation property claimed for the Yee-type (as opposed to Wilson-plaquette) field action. Beyond these algebraic certificates, Section 4.1 reproduces both demonstrations of Shi et al. [2018]—the linear-wave spectra and the laser–plasma slab at $a \simeq 1$ and $a \simeq 16$ —as the physics baseline for everything that follows; all four benchmarks below run in this same code. The reproductions carry their own quantitative criteria, stated here. The linear-wave spectra must place every resolvable branch within the dispersion tolerance of Section 2.3 [Eq. (6)]. For the laser–plasma slab the below-threshold ($a \simeq 1$) pulse-center transit time between fixed stations bracketing the slab must match the geometric-optics prediction $\int dx/v_g(x)$ through the cold density profile to within 5%—a conservative envelope for the windowed centroid-estimator width, the pulse’s 7% spectral bandwidth $[1/(\omega_0 \tau)]$, and edge effects of the finite ramps; above threshold ($a \simeq 16$), where pair creation invalidates the cold refractive index, the pre-stated expectations are qualitative (relativistically induced transparency—a crossing *faster* than cold geometric optics—together with the pair burst, oppositely signed outgoing beams, and spectral broadening of the original Figs. 5–7), with the conservation laws as the quantitative anchor.

2.3 Collective-mode benchmarks and tolerances

Tree-level dispersion. For cold scalar-QED plasma the transverse electromagnetic branch obeys $\omega^2 = \omega_p^2 + k^2$ with $\omega_p^2 = \sum_{\zeta} \omega_{p\zeta}^2$, $\omega_{p\zeta}^2 = q_{\zeta}^2 n_{\zeta 0}/m_{\zeta}$, and the longitudinal (electrostatic) modes

solve $1 + \sum_{\zeta} \chi_{\zeta} = 0$ with the relativistic-quantum susceptibility

$$\chi_{\zeta}(\omega, k) = \frac{\omega_{p\zeta}^2 (k^2 - \omega^2 + 4m_{\zeta}^2)}{(\omega^2 - k^2)^2 - 4m_{\zeta}^2 \omega^2} \quad (4)$$

[Shi et al., 2018, Eliasson and Shukla, 2011, Shi et al., 2016, Hines and Frankel, 1978, Kowalenko et al., 1985]. For a single species, $1 + \chi = 0$ is quadratic in ω^2 : exactly two branches exist (a Langmuir-like branch from ω_p and a pair branch from $2m$), while the acoustic branch of the warm theory collapses onto $\omega = 0$ in the cold limit. For two species Eq. (4) gives a quartic in ω^2 with four branches: a Langmuir-like branch, two pair branches approaching $2m_e$ and $2m_i$ at $k \rightarrow 0$, and a gapless branch absent in cold classical two-fluid theory. The gapless branch is enabled entirely by the $4m_{\zeta}^2$ quantum-recoil terms: seeking $\omega \propto k^2$ and balancing the divergent parts of the two susceptibilities gives the long-wavelength law

$$\omega \rightarrow k^2 \sqrt{\frac{m_e^2 \omega_{pe}^2 + m_i^2 \omega_{pi}^2}{4 m_e^2 m_i^2 (\omega_{pe}^2 + \omega_{pi}^2)}} = \frac{k^2}{2\sqrt{m_e m_i}}, \quad (5)$$

the last equality holding for equal densities of singly charged species, where all dependence on the plasma frequencies cancels: with $\hbar$ restored, $\omega = \hbar k^2 / (2\sqrt{m_e m_i})$ is a single-particle recoil frequency for the geometric-mean mass, a dispersion with no classical limit. (We verified Eq. (5) against numerical roots of the quartic to better than 3×10^{-5} relative at $k = 0.0125 m$, across mass ratios and density mixes.) We evaluate branch frequencies by root-finding on the polynomial form of $1 + \sum_{\zeta} \chi_{\zeta}$ at each k ; no dispersion values are hard-coded.

Prediction and tolerance. The two-species benchmark (S1; "electrons" $q = +0.3$, $m_e = 1$ and "ions" $q = -0.3$, $m_i = 4$, equal densities, $\omega_{pe} = 0.6 m_e$, $\omega_{pi} = 0.3 m_e$, dynamically neutral with no static background) must reproduce every resolvable branch at the wavenumbers listed in Section 3 to within

$$|\omega_{\text{sim}} - \omega_{\text{th}}| \leq \delta_{\omega} = 0.05 m_e. \quad (6)$$

This tolerance is justified before the fact by the three contributions we can bound: the spectral bin width $2\pi/T = 0.0383 m$ of the analysis window (peaks are refined by log-parabolic interpolation, worth a factor of several below one bin), the discretization shift measured by repeating the run at $m\Delta x \in \{0.08, 0.04, 0.02\}$ (observed below $0.02 m$ everywhere, Section 4.2), and the ensemble ridge noise (sub-bin for 100 members). The same tolerance, for the same reasons, is applied to the single-species verification spectra (Section 4.1). The primary diagnostic exercised by S1 is *dispersion fidelity*: an error in any of the mass, hopping, or current terms shifts branch frequencies at leading order, and the pair branches specifically probe the interference of positive- and negative-frequency components that has no classical-plasma analogue.

2.4 Superradiant Klein scattering: prediction and gauge twin

A charged wave packet with central energy E_0 in the Klein zone $m < E_0 < U_0 - m$ of a smooth electrostatic barrier $U(x) = \frac{U_0}{2} [1 + \tanh(x/\ell)]$ scatters superradiantly: the reflected particle current exceeds the incident one, the excess being balanced by an antiparticle current transmitted into the barrier [Klein, 1929, Manogue, 1988, Calogeracos and Dombey, 1999, Hansen and Ravndal, 1981, Dombey and Calogeracos, 1999]. For each incident energy the mode equation $\chi'' + [(E - U(x))^2 - m^2]\chi = 0$ is integrated numerically from the transmitted side (selecting the group-velocity branch $p_t < 0$ inside the Klein zone) and decomposed at the incident side into incoming and reflected amplitudes, yielding $R(E)$ with the Wronskian identity $R + T_{\text{flux}} = 1$, $T_{\text{flux}} < 0$, satisfied to 3×10^{-10} across the packet (the level of the 10^{-10} integration tolerance); the $\ell \rightarrow 0$ limit reproduces the textbook sharp-step coefficient, and the barrier is charge-conjugation symmetric about $E = U_0/2$. The lattice benchmark (S2) prepares a positive-frequency packet $|a_k|^2 \propto \exp[-(k - k_0)^2 / 2\sigma_k^2]$ and measures the final charge partition; the prediction, stated in advance, is the number-weighted average

$$R_{\text{pred}} = \frac{\sum_k |a_k|^2 R(E_k)}{\sum_k |a_k|^2} = 1.8797, \quad |R_{\text{sim}} - R_{\text{pred}}| \leq \delta_R = 0.01 \cdot R_{\text{pred}}, \quad (7)$$

with the tolerance justified by the quadrature of: the packet-width systematic (the weighted spread of $R(E)$ across the packet is 0.011, and the number-weighted mean absorbs it at first order), the

residual charge left near the barrier at measurement time (bounded in the runs below 1.9×10^{-16} of the incident charge), the measured resolution dependence across $m\Delta x \in \{0.08, 0.04, 0.02\}$, and the measured nonlinearity upon doubling the packet amplitude.

Realizing a *single* supercritical step on a periodic lattice is itself instructive. A static potential column must close somewhere on a torus, and any closure at finite distance forms a supercritical cavity: for bosons, with no Pauli blocking, such a cavity is prone to the Schiff–Snyder–Weinberg-type charged-vacuum instability [Schiff et al., 1940, Klein and Rafelski, 1975, Rafelski et al., 1978, Dombey and Calogeracos, 1999], which would contaminate a steady scattering measurement, so closed-column designs are rejected at the outset. The clean realization notes that while the step *potential* is not single-valued on the torus, its *field* sheet $E_x^{\text{ext}} = +\partial_x U/q$ is: it is localized at the step, and in the temporal gauge it is carried by the linearly growing vector potential $a_1^{\text{ext}}(x, t) = t \partial_x(U/q)$, the exact discrete gauge image of the static potential, with no compensating down-step anywhere. (The sign is fixed by the charge convention of Section 2.1: the incident $e^{-i\omega t}$ branch carries physical charge $-q$ and sees potential energy $-q a_0$, so the $+U$ barrier is the gauge image of $a_0^{\text{ext}} = -U/q$.) The background then carries a growing holonomy—an EMF whose work per loop on the incident branch is $-U_0$, the same holonomy physics that Section 2.6 studies dynamically—and, because the ramp is explicitly time dependent, the temporal-gauge discrete energy is *not* conserved in this benchmark: superradiance extracts energy from the source maintaining the step. The pre-stated conservation diagnostics are therefore total charge (exact by construction) and the work–energy ledger $\Delta\mathcal{U}(t) = W_{\text{ext}}(t) = \int^t dt' \int dx j_x E_x^{\text{ext}}$, whose residual must sit at the discretization order of the energy diagnostic, as in the driven benchmark of Section 2.6. Superradiance is a demanding test of the scheme’s charge book-keeping (reflected charge is created, not merely redirected), but the designated primary diagnostic of this benchmark is *gauge invariance*: an exact discrete gauge transformation with a periodic gauge function $\alpha_s^n = -t_n W_s$ (here W a sine of amplitude $2U_0/q$ scaled by 0.3) maps the field-sheet background onto a mixed representation with a static $a_0^{\text{ext}} = -W$ and a modified ramp. Because Eqs. (2)–(3) are exactly U(1) covariant, *every* gauge-invariant observable of the twin runs must coincide to floating-point accuracy; any violation exposes a broken Wilson line. This is a far sharper probe of the gauge sector than statistical agreement between gauges.

2.5 Spontaneous pair creation: failure criterion and quantum-seeded extension

Exact reference. A spatially uniform longitudinal pulse $E_x(t) = E_0 \text{sech}^2[(t - t_c)/\tau]$ (Sauter pulse), represented by $a_1^{\text{ext}}(t) = E_0 \tau [\tanh((t - t_c)/\tau) + \tanh(t_c/\tau)]$, creates pairs from the scalar vacuum with a mode spectrum known exactly: each canonical wavenumber k evolves independently through $\ddot{\chi}_k + [m^2 + (k - qa_1^{\text{ext}}(t))^2]\chi_k = 0$, and the out-state pair occupation is $n_k = |\beta_k|^2$ with the Bogoliubov coefficient obtained by projecting the in-vacuum solution onto the out-basis. We compute $|\beta_k|^2$ two ways: by direct integration of the mode equation (adaptive eighth-order integration, Bogoliubov normalization $|\alpha_k|^2 - |\beta_k|^2 = 1$ maintained to 1.1×10^{-9}), and from the closed-form Narozhnyi–Nikishov-type expression for the Sauter pulse [Narozhnyi and Nikishov, 1970, Gavrilov and Gitman, 1996, Dunne, 2005],

$$|\beta_k|^2 = \frac{\cosh 2\pi\mu + \cosh \pi(\nu_+ - \nu_-)}{2 \sinh(\pi\nu_-) \sinh(\pi\nu_+)}, \quad \nu_{\pm} = \tau \omega_{\pm}, \quad \mu = \sqrt{(qE_0\tau^2)^2 - \frac{1}{4}}, \quad (8)$$

with $\omega_{\pm}$ the asymptotic mode frequencies before/after the pulse; the two references agree to 2.2×10^{-4} (relative to the spectral peak), the residual being the finite pulse-window truncation; for broader context on exact and approximate strong-field pair-creation methods see [Aleksandrov et al., 2017, Fedotov et al., 2023]. In the locally-constant-field regime realized here ($2qE_0\tau^2 \gg 1$) the peak of Eq. (8) approaches the Schwinger factor $\exp(-\pi m^2/qE_0)$ [Schwinger, 1951, Sauter, 1931, Heisenberg and Euler, 1936].

Failure criterion (stated before results). The benchmark applies the pulse to a *field-free* region and measures the produced spectrum by projecting the final field onto lattice out-modes. The scheme fails the benchmark if the yield is not the QFT spectrum $n_k = |\beta_k|^2$. Concretely we test the initialization family in which every mode is seeded with mean occupation $\varepsilon/2$ (complex-Gaussian amplitudes; $\varepsilon = 0$ reproduces the vacuum-region initialization of Shi et al. [2018], $\phi = 0$ plus white-noise A_e), and we state in advance the linear-response law that classical evolution must satisfy: the measured produced spectrum will be $n_k(\varepsilon) = \varepsilon |\beta_k|^2$, vanishing with ε with no ε -independent floor, and exactly zero (to machine precision) at $\varepsilon = 0$ at *every* lattice resolution. Observing this law constitutes the

failure: the classical-statistical scheme contains no scale that fixes ε , so as specified it cannot produce spontaneous pairs, in sharp contrast to the exact result. The fundamental assumption violated is the high-occupation premise of the classical-statistical truncation [Aarts and Berges, 2002, Berges et al., 2014]: spontaneous emission is an $\mathcal{O}(\hbar)$ commutator effect invisible to classical transport of occupations, and $\phi = 0$ is an exact fixed point of any classical charged-scalar flow in any background. The primary diagnostic is the *nontrivial ϕ -A coupling*: the entire observable is generated by the gauge field acting on the matter sector.

Quantum-seeded extension. The extension replaces the initial ensemble, not the action: the discrete equations of motion and all conservation properties of Section 2.1 are unchanged, realization by realization. On the lattice, mode k of the leapfrog evolution oscillates at ω_k given by $4 \sin^2(\omega_k \Delta t / 2) / \Delta t^2 = m^2 + (2/\Delta x)^2 \sin^2[(k - qa_1^{\text{ext}})\Delta x / 2]$, and the symplectically normalized single-particle amplitude carries the leapfrog Wronskian $W_k = \sin(\omega_k \Delta t) / \Delta t$. We seed

$$\hat{\phi}_k(t_0) = \frac{a_k + \bar{b}_{-k}}{\sqrt{2W_k L}}, \quad \hat{\phi}_k(t_0 + \Delta t) = \frac{a_k e^{-i\omega_k \Delta t} + \bar{b}_{-k} e^{+i\omega_k \Delta t}}{\sqrt{2W_k L}}, \quad (9)$$

with $\langle |a_k|^2 \rangle = \langle |b_k|^2 \rangle = \varepsilon/2$ and $\varepsilon = 1$ the quantum vacuum [Polkovnikov, 2003, Gelis and Tanji, 2013, Kasper et al., 2014, Hebenstreit et al., 2013a,b, Tanji, 2009]; occupations are read out by the exact inverse projection at the measurement-time background, and the seeded input $\varepsilon/2$ is subtracted (vacuum subtraction). With this normalization the seeded ensemble is the exact Wigner distribution of the lattice ground state: under free evolution the measured occupations remain $\varepsilon/2$ identically—verified over the full production window with the pulse off, where the largest vacuum-subtracted occupation over 2000 members is 3.8×10^{-2} , i.e. 3.7 standard errors (the expected extreme of 1280 Gaussian modes), and the integrated density is consistent with zero at 1.0σ —and for the quenched (external-field) problem—which is linear—the classical-statistical average reproduces symmetric-ordered two-point functions exactly, so the only sources of deviation from Eq. (8) are lattice discretization and ensemble sampling. The verification tolerance, stated in advance, is

$$|n_k^{\text{sim}} - |\beta_k|^2| \leq \max(3\sigma_k^{\text{ens}}, \delta_k^{\text{disc}}) \quad \text{for all } k \text{ in the production band}, \quad (10)$$

where σ_k^{ens} is the measured per-mode ensemble standard error and δ_k^{disc} the resolution envelope measured across $m\Delta x \in \{0.08, 0.04, 0.02\}$; the production band is defined as the modes where the reference spectrum exceeds 5% of its peak; the integrated density must agree within the same combined uncertainties. For the field-strength scan the peak occupation is estimated as the mean over a $\pm 0.3m$ window centred on the predicted spectral peak, with the reference averaged over the same window: a maximum over modes would be biased upward by extreme-value sampling noise at finite ensembles, while the window mean is unbiased and carries a propagated ensemble error. Charge conservation remains exact per realization (verified at machine precision); the matter energy is not conserved in the driven background—the pulse does work on every mode—so the energy diagnostic is again a work–energy ledger, $\Delta \mathcal{U}_\phi(t) = \int^t dt' E_{\text{ext}}(t') \int dx j_x$, which must close at the discretization order; the Gauss constraint is respected by construction in the quenched runs (frozen dynamical gauge sector, the regime of the analytic reference) and solved explicitly at initialization in the fully self-consistent runs, for which we quantify the backreaction field $\langle \delta E_x \rangle / E_0$. What the extension *relaxes* is neither symmetry nor conservation but a premise of the sampling: initial data are no longer classical fields of definite amplitude but samples of the vacuum Wigner functional, and observables acquire the associated vacuum subtraction; the cost of carrying the half-quantum in every mode is quantified in Section 5.

2.6 Secular acceleration: failure criterion and holonomy-resolved extension

Lattice prediction and failure criterion (stated before results). Apply a constant force $F = qE_x$ to a dilute cold condensate (comoving density n_c , negligible backreaction, verified) via the uniform external potential $a_1^{\text{ext}}(t) = E_x t$. In the continuum the current is monotone and saturates at the speed of light,

$$j_{\text{cont}}(t) = q n_c v(t), \quad v(t) = \frac{Ft}{\sqrt{m^2 + F^2 t^2}} \xrightarrow{t \rightarrow \infty} 1, \quad (11)$$

(adiabatic form; our quantitative reference integrates the $k = 0$ mode equation exactly, and the two agree to 6.3×10^{-4}). On the lattice the same mode obeys the exact single-mode recurrence with

kinetic energy $(2/\Delta x)^2 \sin^2(qE_x t \Delta x/2)$: the kinetic momentum $p(t) = Ft$ is dragged across the compact Brillouin zone, and the current $j \propto \sin(p \Delta x)$ *reverses* at

$$t_{\text{rev}} = \frac{\pi}{qE_x \Delta x}, \quad \text{with pre-reversal error growth} \quad \frac{|j - j_{\text{cont}}|}{\max j_{\text{cont}}} \simeq \frac{(qE_x t \Delta x)^2}{6}. \quad (12)$$

The failure criterion, stated in advance: a *deep reversal*—the current crosses zero downward after having exceeded $\frac{1}{2}qn_c$ and subsequently drops below $-\frac{1}{4}qn_c$ —occurring at finite time at every resolution, with onset scaling as $1/\Delta x$ per Eq. (12). Continuum dynamics never triggers this criterion. Because the reversal time *diverges* as $\Delta x \rightarrow 0$ while always remaining finite, the scheme converges on any fixed time window yet fails uniformly in time: for sustained acceleration the resolution requirement $p_{\text{max}} \Delta x < \pi$ is set by the momentum *reached*, not by the initial data—a structural property, not a bug, which we corroborate by (i) machine-precision charge and Gauss conservation throughout the failing runs, (ii) agreement of the full simulation with the exact single-mode lattice recurrence to $\lesssim 2 \times 10^{-3}$, and (iii) the same code reproducing all benchmarks above. The failure saturates the Wilson-line periodicity bound $qA\Delta < 2\pi$ of Shi et al. [2018] at exactly half its naive value: the group velocity turns over at the zone boundary $p \Delta x = \pi$. No gauge choice evades it—the drag rate of the gauge-invariant link phase is fixed by E_x ; in Lorenz gauge a uniform field on a torus still requires a secularly growing vector potential—and this gauge-sector character is why we designate the primary diagnostic of this benchmark as *charge/energy conservation book-keeping*: the work–energy ledger $dU/dt = E_x j L$ closes at $\mathcal{O}((\Omega_{\text{max}} \Delta t)^2)$ while the energy *content* saturates at the band edge instead of growing as $\gamma(t)$, cleanly separating “the integrator is conservative” from “the physics is right.”

Holonomy-resolved extension. On a spatial torus the gauge potential splits into an exact part, a transverse part, and the harmonic (holonomy) mode $\bar{a}_1(t)$ —the spatial mean, which is precisely the component that grows secularly under a sustained field and cannot be gauged away (large gauge transformations shift it by $2\pi/qL$ only). The extension treats matter spectrally and couples the holonomy *noncompactly*:

$$S_{\text{hol}} = \sum_n \Delta t L \left\{ \sum_k \left[\frac{|\hat{\phi}_k^{n+1} - \hat{\phi}_k^n|^2}{\Delta t^2} - (m^2 + (k - q\bar{a}_{1,\text{tot}}^n)^2) |\hat{\phi}_k^n|^2 \right] + \frac{(\bar{e}^{n+1/2})^2}{2} \right\}, \quad (13)$$

with $\bar{e}^{n+1/2} = (\bar{a}_1^{n+1} - \bar{a}_1^n)/\Delta t$ and $\bar{a}_{1,\text{tot}} = \bar{a}_1 + a_1^{\text{ext}}$. Extremizing gives the leapfrog mode equations $\hat{\phi}_k^{n+1} = 2\hat{\phi}_k^n - \hat{\phi}_k^{n-1} - \Delta t^2 [m^2 + (k - q\bar{a}_{1,\text{tot}}^n)^2] \hat{\phi}_k^n$ and the mean Ampère law $\bar{e}^{n+1/2} = \bar{e}^{n-1/2} - \Delta t \bar{j}^n$ with the noncompact current $\bar{j} = -2q \sum_k (k - q\bar{a}_{1,\text{tot}}) |\hat{\phi}_k|^2$. Conservation properties: each mode carries an exactly conserved leapfrog Wronskian, so *total charge is conserved to machine precision mode by mode*; the discrete energy is conserved to $\mathcal{O}(\Delta t^2)$ and exactly at $q = 0$; and the scheme is exactly covariant under large gauge transformations $\bar{a}_1 \rightarrow \bar{a}_1 + 2\pi/qL$, $\hat{\phi}_k \rightarrow \hat{\phi}_{k-2\pi/L}$ (verified at machine precision), mirroring continuum torus QED, in which the holonomy is a genuine global degree of freedom [Manton, 1985] and kinetic momenta $k - q\bar{a}_1$ are unbounded even though canonical modes are discrete. What the extension *relaxes* is spatial locality of the matter–holonomy coupling: charge conservation becomes mode-local instead of site-local, and the momentum operator is spectral (all-to-all in x). We regard this as the honest price of the physics: the holonomy is itself a global (cohomological) degree of freedom, so no strictly local coupling can represent its unbounded drift. The extension also converts the spatial horizon into a time-resolution requirement, $\Delta t \lesssim 1/\omega_{\text{max}}(T) = 1/\sqrt{m^2 + (FT)^2}$ —cost growing linearly with the target momentum rather than an impossibility—which we verify by a Δt -convergence study. In this work the extension is implemented and verified for the uniform-sector problem that exhibits the failure (spatially uniform accelerating fields with an arbitrary spectrum of matter modes); coupling the holonomy-resolved matter sector to *inhomogeneous* dynamical gauge fluctuations is formulated by the same split but is left unimplemented, a limitation discussed in Section 5.

2.7 Diagnostic coverage

The four benchmarks were chosen so that their primary diagnostics cover the four distinct failure classes of a lattice gauge-matter code, as argued in each subsection above: dispersion fidelity (S1), exact gauge invariance (S2), nontrivial ϕ – A coupling (B1), and conservation/energy-ledger book-keeping (B2); the two reproductions of Shi et al. [2018] exercise all four in combination but none in

isolation. Every run logs the total charge $Q(t)$, the Gauss residual, and the discrete energy $\mathcal{U}(t)$ —for the externally driven configurations, whose backgrounds do work on the matter, the energy diagnostic is the work–energy ledger defined in their subsections; the table in Section 4.6 reports $|\Delta Q/Q|$ and $|\Delta \mathcal{U}/\mathcal{U}|$ for one representative realization of every benchmark.

3 Experimental Setup

All simulations use $q = 0.3$, electron mass $m = 1$ (natural units), the temporal gauge, periodic boundaries, and the explicit variational update of Section 2.1; the code is a fresh implementation written for this study (Python/NumPy with compiled kernels), with every run emitting a JSON configuration (parameters, seed lists, UTC timestamps, code hash) and raw time series alongside the derived numbers quoted here. Ensemble members are statistically independent (per-member seeds logged) and run in parallel on up to 28 CPU cores.

Verification runs. The linear-wave reproduction follows Shi et al. [2018] Sec. III.1: single species, immobile neutralizing background, $\omega_p = 0.85$ ($n_0 = \omega_p^2 m/q^2 \approx 8.03$), $m\Delta x = 0.04$, $m\Delta t = 0.02$, $L = 512$ points, $T = 1024$ steps, ensemble of 100, gauge links seeded with white noise of standard deviation 10^{-4} , condensate phases random. For quantitative peak extraction the same physics is run in a fixed box $L = 20.48/m$ for $T_{\text{phys}} = 163.84/m$ at $m\Delta x \in \{0.08, 0.04, 0.02\}$ ($\Delta t = \Delta x/2$), giving spectral bins $2\pi/T_{\text{phys}} = 0.0383m$; branch frequencies are read at the box harmonics $k = 2\pi n/L$ with $n \in \{1, 2, 3, 4, 6, 8\}$ ($k/m \approx 0.31\text{--}2.45$) by log-parabolic peak interpolation. The laser-plasma reproduction follows Sec. III.2: slab of peak density $n_0 = 1$ ($\omega_p = 0.3$), plateau $100/m$ wide centered at $x = 400/m$ with Gaussian ramps $\sigma = 20/m$ in a box of length $720/m$; the neutralizing background equals the initial matter density profile; the plasma is phase-mixed without the laser for $400/m\text{--}800/m$ and sampled at 200 random times to build the ensemble; a right-moving pulse $A^y \propto \exp(-\xi^2/2\tau^2) \cos \omega_0 \xi$ with $\omega_0 = 0.7$, $\tau = 20/m$, zero carrier phase, is then added at $x_0 = 200/m$ with $a = qA_{\text{max}}^y/m \in \{1, 16\}$ (the smooth immobile background suppresses trident and Breit–Wheeler channels [Bjorken and Chen, 1967, Breit and Wheeler, 1934], as in the original setup); $m\Delta x = 0.04$, $m\Delta t = 0.005$, $T = 460/m$, ensemble of 200 per intensity, and a 50-member run at $m\Delta x = 0.08$ checks resolution.

Two-species plasma (S1). Species $(q, m) = (+0.3, 1)$ and $(-0.3, 4)$ at equal density $n_0 = 4$ ($\omega_{pe} = 0.6$, $\omega_{pi} = 0.3$), no static background, same noise seeding, box $20.48/m$, $T_{\text{phys}} = 163.84/m$, ensembles of 100 at $m\Delta x \in \{0.08, 0.04, 0.02\}$; peaks are read at the harmonics $n \in \{2, 3, 4, 6, 8, 10\}$ ($k/m \approx 0.61\text{--}3.07$) for all four longitudinal branches and the transverse branch (the gapless branch is archived under the label “acoustic” in the run artifacts).

Klein barrier (S2). Barrier $U_0 = 3m$, $\ell = 0.5/m$, realized as the field sheet of Section 2.4 (no down-step anywhere); packet central energy $E_0 = 1.5m$ ($k_0 = \sqrt{1.25}m$), width $\sigma_k = m/12$, amplitude 3×10^{-4} (self-field effects checked by amplitude doubling), box $x \in [-160, 352]/m$, $T = 180/m$; main run $m\Delta x = 0.04$, $m\Delta t = 0.005$ with an 8-member ensemble jittering the launch position, resolution scan $m\Delta x \in \{0.08, 0.04, 0.02\}$, and the exact gauge twin of Section 2.4. Charge is measured in the regions $x < -5/m$ (reflected), $|x| \leq 5/m$ (barrier), and $x > 5/m$ (transmitted); the work–energy ledger W_{ext} is accumulated every step from the matter current against the static field sheet.

Spontaneous pair creation (B1). Sauter pulse with $m\tau = 4$, centered $t_c = 20/m$, measured at $T = 40/m$; field strengths $qE_0/m^2 \in \{1.0, 1.5, 2.0\}$ (i.e. $E_0/E_S \in \{1, 1.5, 2\}$ with $E_S = m^2/q$ the Schwinger field); box $51.2/m$; main resolution $m\Delta x = 0.04$, $m\Delta t = 0.005$ with scans at $m\Delta x \in \{0.08, 0.02\}$ ($\Delta t = \Delta x/8$); seed families $\varepsilon \in \{0, 0.03, 0.1, 0.3, 1\}$ with 2000 members each (100 for the $\phi = 0$ initialization of Shi et al. [2018], which is exactly deterministic in the matter sector); quenched gauge sector for the analytic-reference comparisons, plus a 500-member fully self-consistent run with the Gauss constraint solved at initialization.

Secular acceleration (B2). Constant force $F = qE_x = 0.05m^2$ ($E_x = E_S/20$, adiabatic and far below the pair-creation threshold) applied to a dilute condensate $n_c = 10^{-8}m^3$ (measured backreaction below 1.8×10^{-5} of E_x); compact-scheme runs at $m\Delta x \in \{0.32, 0.16, 0.08, 0.04\}$ ($\Delta t = \Delta x/4$) to $2.5t_{\text{rev}}(\Delta x)$; holonomy-resolved runs to $T = 2.5t_{\text{rev}}(0.04) \approx 3927/m$ at

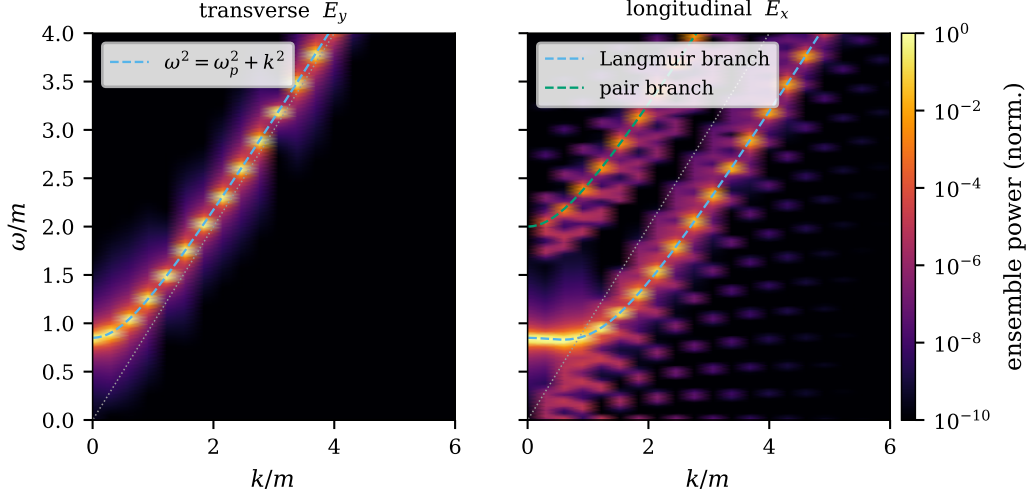


Figure 1: Verification against Shi et al. [2018] Sec. III.1: ensemble power spectra of the transverse (left) and longitudinal (right) electric field for the single-species plasma ($\omega_p = 0.85m$, $m\Delta x = 0.04$, 100 members; shown from the extended window $T_{\text{phys}} = 163.84/m$ used for peak extraction—the run at the original T is in the logs and is visually identical at coarser frequency bins), overlaid with the tree-level dispersion relations (dashed). The longitudinal spectrum shows the Langmuir-like branch emerging from ω_p and the pair branch from $2m$; the dotted gray line is the light cone. In the cold limit the acoustic branch of the warm theory degenerates into the $\omega = 0$ band.

$m\Delta t \in \{0.002, 0.001, 0.0005\}$; a multi-mode variant seeds a Gaussian band of matter modes ($\sigma_k = 2m$) evolved by both schemes against per-mode continuum integrations; and a work–energy ledger run at halved Δt isolates the $\mathcal{O}((\Omega\Delta t)^2)$ bookkeeping term.

4 Experimental Results

4.1 Verification against the original demonstrations

Figure 1 reproduces the linear-wave demonstration: both electromagnetic and electrostatic ensemble spectra ride the tree-level curves across the Brillouin zone, including the pair branch at $\omega \geq 2m$ that distinguishes relativistic-quantum from classical plasmas [Fuda and Furlani, 1982, Kowalenko et al., 1985]. Quantitatively, at the six tabulated wavenumbers ($k/m = 0.31\text{--}2.45$) the extracted peaks deviate from the analytic branches by at most $0.0013m$ (transverse), $0.0012m$ (Langmuir), and $0.0050m$ (pair) at the production resolution, well inside the tolerance (6); the deviations at $m\Delta x \in \{0.08, 0.04, 0.02\}$ (Fig. 9a) confirm they are dominated by spectral estimation rather than discretization. Charge is conserved to 4.9×10^{-15} (relative), the Gauss residual stays below 3.3×10^{-13} , and the discrete energy of the noise-level run is constant to 1.8×10^{-11} .

Figures 2–3 reproduce the laser–plasma demonstration at both intensities with the phase-mixed ensemble of Shi et al. [2018]. All qualitative features of the original Figs. 5–7 appear: ponderomotive snow-plow and wakefield acceleration [Tajima and Dawson, 1979, Pukhov and Meyer-ter Vehn, 2002, Esarey et al., 2009] at $a \simeq 1$ with "electrons" splashed backward from the driven boundaries [McKinstrie and Startsev, 1996, Naumova et al., 2004, Geyko et al., 2009]; at $a \simeq 16$, pair creation concentrated where the laser overlaps plasma waves, oppositely signed charge beams leaving the slab, strong depletion of the electromagnetic energy, and forward Raman sidebands at $\omega_0 + n\omega_p$ below threshold [Kruer, 1988] versus broadband spectra above, the frequency content shifting as expected from wave-action exchange with the plasma [Wilks et al., 1988, Dodin and Fisch, 2010]. Quantitatively: at $a \simeq 1$ the pulse-center transit time between fixed stations bracketing the slab (windowed centroid of the electromagnetic energy density) is $193.5/m$ against the geometric-optics prediction $195.6/m$ —a deviation of 1.1%, within the pre-stated 5% tolerance; expressed as the slowdown relative to vacuum flight, $13.5/m$ is measured versus $15.6/m$ predicted, the residual

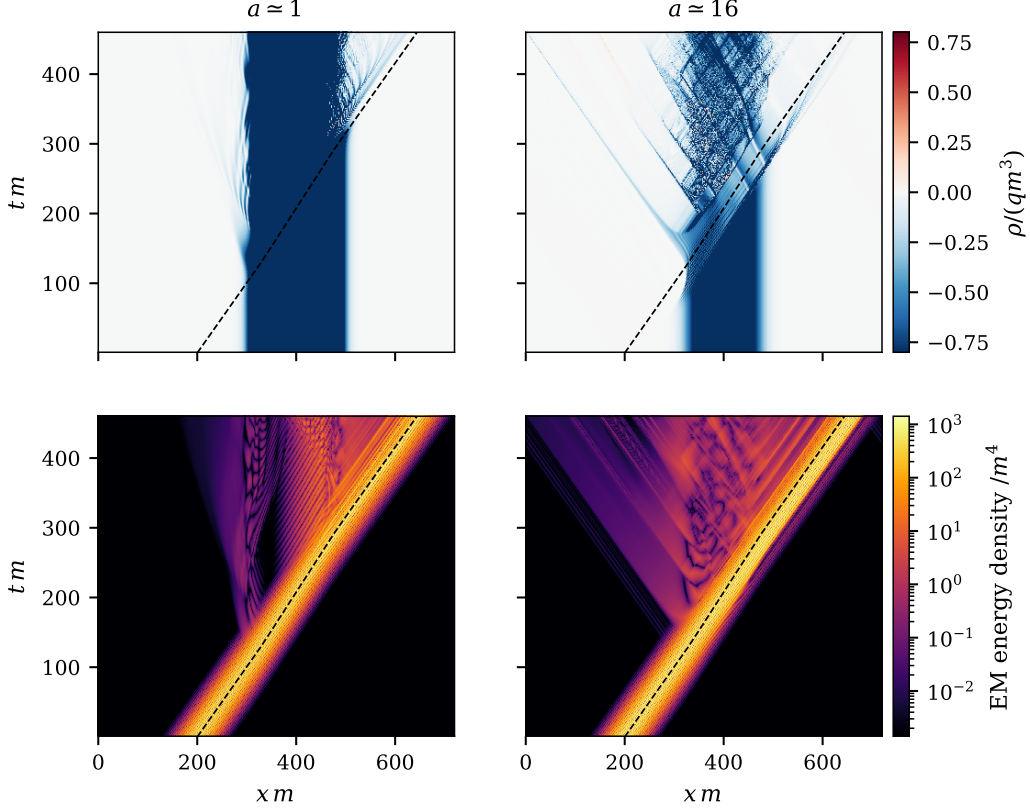


Figure 2: Verification against Shi et al. [2018] Sec. III.2: ensemble-mean charge density of the matter field (top; blue: net "electron" excess, red: net "positron" excess) and electromagnetic energy density (bottom, log scale) for the gamma-ray laser ($\omega_0 = 0.7m$) at $a \simeq 1$ (left) and $a \simeq 16$ (right). Black dashed: geometric-optics pulse-center trajectory through the slab. At $a \simeq 1$ the laser drives a wakefield and splashes "electrons" from the boundaries; at $a \simeq 16$ ($E_x \approx 5E_S$) copious pairs are produced where the laser and its backscatter interact with plasma waves, and beams of both signs leave the slab.

consistent with the pulse's 7% spectral bandwidth and the centroid estimator. At $a \simeq 16$ the pulse crosses 5.9% *faster* than cold-slab geometric optics—the expected relativistic transparency of a strongly driven, pair-producing slab, for which the cold refractive index no longer applies. Total charge is conserved to 6.0×10^{-14} ($a \simeq 1$) and 3.9×10^{-14} ($a \simeq 16$); the total energy is conserved to 1.2×10^{-6} and 3.5×10^{-4} respectively, the same orders as the 6 ppm and 0.2% figures reported for the original implementation, confirming the $\mathcal{O}(q\Delta^2)$ scaling of the energy defect at strong fields. Matter absorbs 6.2% of the pulse energy below threshold and 8% above; in absolute terms the matter energy grows 14-fold above threshold, and the charge-density map develops the positron-rich filaments and oppositely signed beams of the original Fig. 5(b)—net *positive* charge densities that cannot occur without pair creation. One quantitative aspect is weaker in our ensemble than the original Fig. 7 suggests: the above-threshold depletion of the electromagnetic energy, which the original account attributes dominantly to pairs created where the *backscattered* wave beats with plasma waves, is exponentially sensitive to the plasma-fluctuation level seeding the backscatter, and our independently phase-mixed ensemble evidently carries a weaker microscopic seed at the same macroscopic profile; Section 5 connects this sensitivity to the occupation dichotomy that the vacuum benchmark isolates. A 50-member run at half resolution reproduces the transit time within 0.03% and the below-threshold energy partition to 0.04%, confirming convergence of the observables used here.

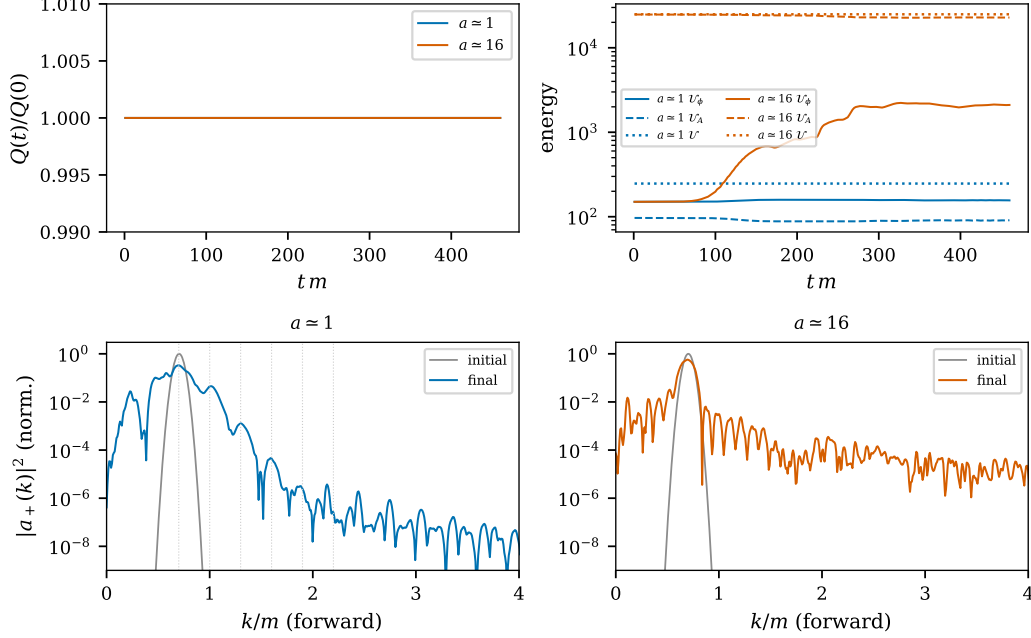


Figure 3: Laser-plasma verification, conservation and spectra. (a) Total charge (normalized to the background ion charge) and (b) field/matter/total energies for single realizations at $a \simeq 1$ and $a \simeq 16$; charge is flat to machine precision, total energy to the $\mathcal{O}(q\Delta^2)$ level quoted in the text. (c) Forward ($k > 0$) transverse spectra before and after transit at $a \simeq 1$, showing the Raman ladder at $\omega_0 + n\omega_p$; (d) the same at $a \simeq 16$, showing the pair-creation-induced broadening.

4.2 Two-species pair-ion plasma: four-branch spectrum

Figure 4 shows the central scope result: with both species dynamical and no static background, the longitudinal ensemble spectrum displays *four* branches exactly where the two-species susceptibility puts them. Peak extraction at the six tabulated wavenumbers gives maximum deviations of $0.0018 m$ (transverse), $0.0013 m$ (gapless branch), $0.0008 m$ (Langmuir-like), and $0.0016 m$ (electron-pair branch), each within the tolerance (6) at every resolution studied (Fig. 9b). The ion-pair branch near $2m_i = 8m_e$ is excited by the noise seeding at the 10^{-5} level of the longitudinal power and is visible in the spectrum exactly on its predicted curve (Fig. 4); its peak is resolvable against the noise floor at $k/m_e \gtrsim 0.61$, where it deviates from the prediction by at most $0.003 m$. The gapless branch merits emphasis: a cold classical two-fluid plasma supports no propagating electrostatic mode below the plasma frequency, yet here a branch rises out of the origin with the purely quantum dispersion of Eq. (5). At the smallest wavenumber where the branch clears the spectral resolution, $k = 0.92 m$, the measured frequency $0.1774 m$ sits on the quartic prediction ($0.1787 m$), which at this k has already bent away from the pure-recoil asymptote $k^2/(2\sqrt{m_e m_i}) = 0.2118 m$ —the spectra thus resolve the crossover out of the recoil regime, a two-species relativistic-quantum effect that the original single-species demonstrations could not exhibit. Charge is conserved to 4.7×10^{-15} and energy to 1.4×10^{-11} .

4.3 Superradiant Klein scattering

Two observations frame this benchmark. The first is a verification lesson. Our first production run of the field-sheet geometry realized, through a sign slip in the branch-potential mapping (the $e^{-i\omega t}$ branch sees $-q a_0$; Section 2.4), the *charge-conjugate* configuration: a $-U$ well for the incident branch instead of the $+U$ barrier. The pre-stated prediction caught the slip immediately and quantitatively: the measured $R = 0.03667$ disagreed with the superradiant prediction by a factor of 51 but agreed with the mode-matching coefficient recomputed for $U_0 \rightarrow -U_0$, $\langle R \rangle_{\text{well}} = 0.03667$, to 1.3×10^{-4} (relative)—at all three lattice resolutions. The wrong-sign run is thus itself a stringent, if unplanned, validation of the scattering machinery on the conjugate problem, and it makes concrete

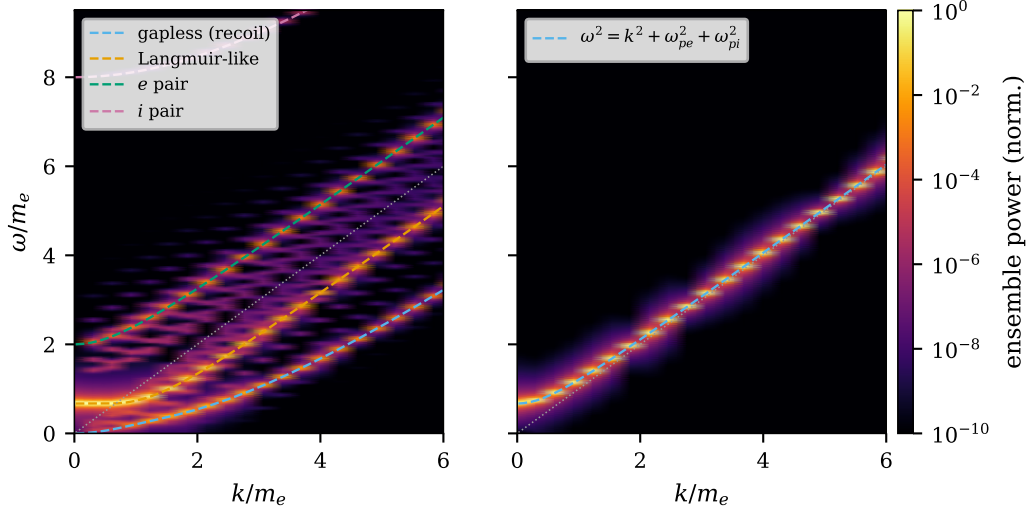


Figure 4: Two-species plasma with dynamical ions ($m_i = 4m_e$, $\omega_{pe} = 0.6$, $\omega_{pi} = 0.3$). Left: longitudinal ensemble spectrum with the four predicted branches of $1 + \chi_e + \chi_i = 0$ overlaid (gapless recoil branch, Langmuir-like, electron-pair, ion-pair); right: transverse spectrum with $\omega^2 = k^2 + \omega_{pe}^2 + \omega_{pi}^2$. The gapless branch is a purely quantum cold mode: it exists only through the $4m_\zeta^2$ recoil terms in Eq. (4).

that supercritical *steps* and their conjugate *wells* are different physical objects for charged scalars: the well transmits 96% of the incident charge with ordinary $R + T = 1$ book-keeping and creates no pairs. Both runs, executed by identical machinery, are archived; all numbers below are from the corrected-sign runs.

With the sign corrected, the scattered state (Fig. 5) shows textbook superradiance realized self-consistently on the lattice: the reflected charge is $R = 1.8797$ times the incident charge, against the pre-stated prediction 1.8797—agreement to 0.002%, inside the 1% tolerance of Eq. (7)—while the transmitted region acquires the balancing *opposite* charge ($Q_{\text{right}}/Q_{\text{inc}} = -0.880$, with $1 - R = -0.880$) and the *total* charge is flat to 9.4×10^{-14} . The eight launch-position members agree among themselves to 2.0×10^{-9} : linear scattering off a static profile is launch-invariant, so the jitter ensemble bounds boundary and wrap-around artifacts rather than sampling noise. Resolution halving and doubling shift R by at most 1.2×10^{-4} (Fig. 5b), amplitude doubling by 1.6×10^{-7} , and the charge left near the barrier at measurement time is below 1.9×10^{-16} of the incident charge—together justifying the stated tolerance a posteriori. Energy is not conserved here, nor should it be: the matter–field energy of the scattering realization grows by a factor 2.76 as superradiance draws energy from the source maintaining the step, and the pre-stated work–energy ledger of Section 2.4 closes to 7.9×10^{-5} of the peak energy (Fig. 5c), with total charge exact throughout. The gauge twin—the same physics presented to the code in a mixed representation with a static electrostatic component and a modified ramp—reproduces the charge-density field and every derived observable to 1.8×10^{-14} (relative to the density scale) and the reflection coefficient to $|R_A - R_B| = 1.0 \times 10^{-13}$: the discrete Wilson-line machinery is exactly gauge covariant in a background whose link phases wrap by tens of radians over the run. Superradiance is stimulated pair creation at a barrier, and the lattice scheme—with no vacuum seeding—captures it quantitatively because the incident packet occupies the relevant modes; this is the positive half of the occupation dichotomy sharpened next.

4.4 Spontaneous pair creation requires the quantum half

Figure 6 presents the first boundary result exactly as pre-registered in Section 2.5. With the initialization of the original scheme ($\phi = 0$ in the field-free region, white-noise gauge links), the supercritical pulse produces *nothing*: the matter field remains identically zero to machine precision at every resolution ($\max_j |\phi_j| = 0$ after the pulse), because $\phi = 0$ is an exact fixed point of the classical flow—while the exact spectrum integrates to 0.115 pairs per unit length. Seeding the ensemble

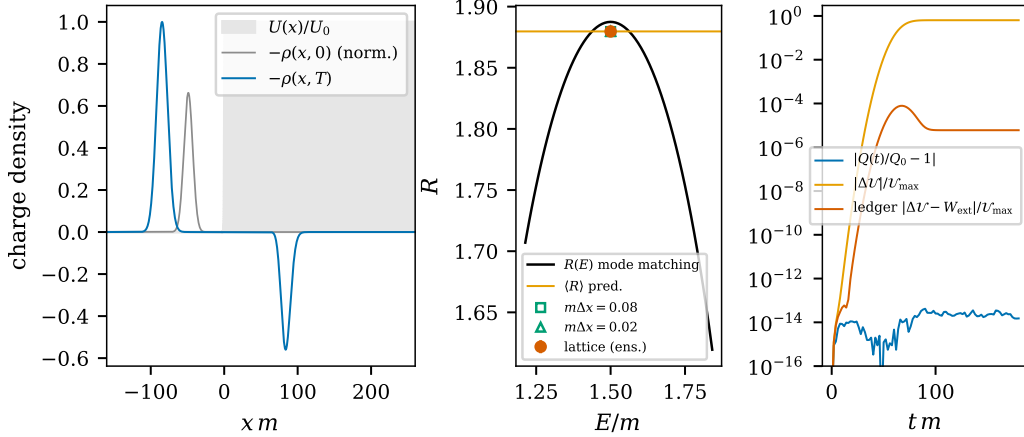


Figure 5: Superradiant scattering off a supercritical Sauter barrier ($U_0 = 3m$, $\ell = 0.5/m$). (a) Matter charge density before (gray) and after (blue) scattering for one realization; the barrier profile is shaded. The reflected packet carries *more* negative charge than was incident; the balancing positive (antiparticle) charge is transmitted past the step. (b) Reflection coefficient: mode-matching prediction $R(E)$ across the packet (curve), number-weighted prediction (line), and lattice measurements at three resolutions (points; the launch-jitter spread is far below visibility). (c) Conservation for one realization: total charge is flat at machine precision while the total energy grows by the superradiant gain; the work–energy ledger residual $|\Delta\mathcal{U} - W_{\text{ext}}|/U_{\max}$ stays at the discretization level.

at mean occupation $\varepsilon/2$ per mode produces precisely the linear law announced in advance: the measured spectra collapse onto $\varepsilon|\beta_k|^2$ (Fig. 6a), and the integrated yield fits $n(\varepsilon) = 0.993\varepsilon n_{\text{QFT}}$ with intercept $-1.2 \times 10^{-15} n_{\text{QFT}}$ (Fig. 6b). The machine-zero intercept is exact by construction, and instructively so: the ε families share common random seeds and the quenched evolution is linear in ϕ , so $n_k = \varepsilon|\beta_k|^2$ holds realization by realization, not merely in the ensemble mean, while the slope’s deviation from unity (0.993) measures the lattice discretization of $|\beta_k|^2$ itself. There is no ε -independent floor, so no classical initialization—which contains no scale to fix ε —can produce the spontaneous yield. This is the failure of the high-occupation premise, isolated quantitatively.

The quantum-seeded extension ($\varepsilon = 1$) converts the same code into a precision instrument for spontaneous production. At the production resolution the measured spectrum tracks the exact $|\beta_k|^2$ across the entire production band (n_k up to 0.121 at the peak) within the pre-stated tolerance (10): the maximum deviation is 0.0366 against a bound of $\max(3\sigma^{\text{ens}}, \delta^{\text{disc}}) = 0.0455$, and the integrated density agrees with the exact reference to 0.7% (ensemble error 1.9%). Particle and antiparticle spectra agree mode by mode (n_k^a vs n_{-k}^b within 0.0568; the largest normalized difference across the 77-mode production band is 3.4 combined standard errors, consistent with ensemble statistics), the projection is consistent with the seeds at $t=0$ to 4.3×10^{-13} , charge is conserved per realization to 1.8×10^{-14} , and the quenched work–energy ledger of a representative realization closes to 4.9×10^{-3} of the peak external work (1.6×10^{-5} of the vacuum-background energy): the pulse deposits net work -127.8 into this realization’s matter field against a measured energy change of -128.5 . Halving and doubling the resolution moves the spectrum by less than 0.0455 at every mode (Fig. 6c); the shift is itself statistically consistent with the independent ensembles drawn at each resolution (largest normalized difference 2.4 combined standard errors), so discretization is subdominant to sampling at these ensemble sizes. Scanning the field over $E_0/E_S \in \{1, 1.5, 2\}$ confirms the Schwinger exponent: peak-window occupations $\{0.040 \pm 0.005, 0.121 \pm 0.006, 0.202 \pm 0.007\}$ against the identically windowed exact values $\{0.043, 0.121, 0.204\}$ (Fig. 6d)—each within one standard error across a factor of two in field strength. Finally, restoring the dynamical gauge sector (Gauss-consistent initialization, backreaction on) does not perturbatively correct the quenched result—it leaves the quenched regime altogether: the seeded vacuum and the created pairs discharge the applied pulse, generating a mean field $\langle \delta E_x \rangle / E_0 = 4.1 \times 10^{-1}$ by the end of the run, and occupations projected against the uniform-background out-basis cease to be meaningful there (the apparent integrated yield inflates 33-fold, an artifact of projecting onto a basis the dynamical background has rotated away).

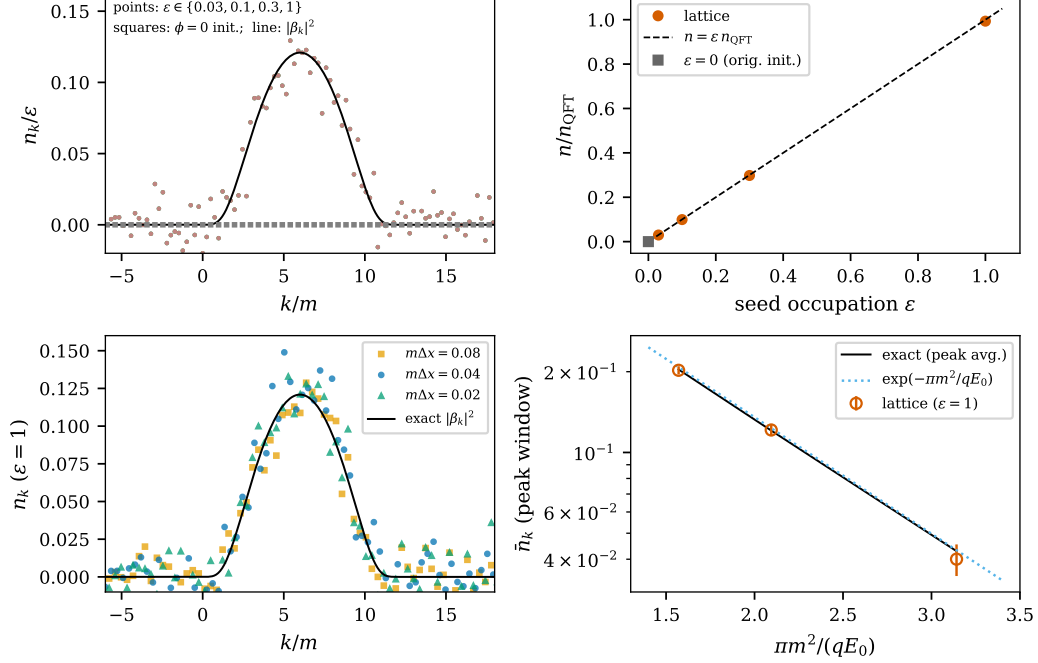


Figure 6: Spontaneous Sauter–Schwinger pair creation from a field-free region ($E_0 = 1.5 E_S$, $m\tau = 4$). (a) Produced spectrum at seed occupations $\epsilon/2$ per mode: measured n_k (points, ensembles of 2000) versus $\epsilon |\beta_k|^2$ (curves). The $\phi = 0$ initialization of Shi et al. [2018] ($\epsilon = 0$, squares) yields identically zero. (b) Integrated yield versus ϵ : the measured line through the origin confirms $n(\epsilon) = \epsilon n_{\text{QFT}}$ with no classical floor; only $\epsilon = 1$ (the Wigner half-quantum) reproduces the exact yield. (c) Quantum-seeded spectrum versus the exact Bogoliubov spectrum at three resolutions. (d) Peak-window occupation ($\pm 0.3 m$ about the predicted peak, with ensemble errors) versus E_S/E_0 : the measured exponent matches the Schwinger factor $\exp(-\pi m^2 / q E_0)$ across a factor of two in field.

The quenched sector is therefore the regime of quantitative validity for the $\epsilon = 1$ comparison; the strength of the self-consistent response is the vacuum-load backreaction discussed in Section 5, not a small correction to it.

4.5 Secular acceleration: the compact-momentum horizon and its removal

Figure 7 shows the second boundary result, again exactly as pre-registered. At every resolution the compact-scheme current rises along the continuum curve, peels away once the accumulated momentum approaches the zone boundary, and then *reverses*—the deep-reversal criterion of Section 2.6 fires at $t_{\text{rev}}^{\text{sim}}/t_{\text{rev}}^{\text{pred}} = \{1.0000, 1.0000, 1.0000, 1.0000\}$ for $m\Delta x = \{0.32, 0.16, 0.08, 0.04\}$, confirming the parameter-free prediction $t_{\text{rev}} = \pi / (q E_x \Delta x)$ to a fraction of a percent across an eightfold range of resolutions. Before reversal the relative error grows as the derived law $(q E_x t \Delta x)^2 / 6$ (Fig. 7c): the four resolutions collapse onto a single curve in t/t_{rev} , demonstrating that refining the lattice rescales but never removes the horizon. Throughout the failing runs the machinery is exact where it should be: charge is conserved to 1.7×10^{-13} , the Gauss residual stays below 6.3×10^{-22} , the full simulation agrees with the exact single-mode lattice recurrence to 1.9×10^{-3} , backreaction is bounded by 1.8×10^{-5} , and the work–energy ledger closes to 4.4×10^{-2} of the peak energy—a residual that halves twice when Δt is halved (1.1×10^{-2} at $\Delta t \rightarrow \Delta t/2$), identifying it as $\mathcal{O}((\Omega \Delta t)^2)$ diagnostic bookkeeping rather than drift. The failure is therefore structural: a Bloch oscillation of the compact matter dispersion, not an error of the integrator.

The holonomy-resolved extension removes the horizon completely. With the noncompact mean-field coupling of Eq. (13), the condensate current follows the continuum reference across the full window $T = 2.5 t_{\text{rev}}(0.04) \approx 3927/m$ —final kinetic momentum $p = FT \approx 196 m$, i.e. $\gamma \approx 196$ —with

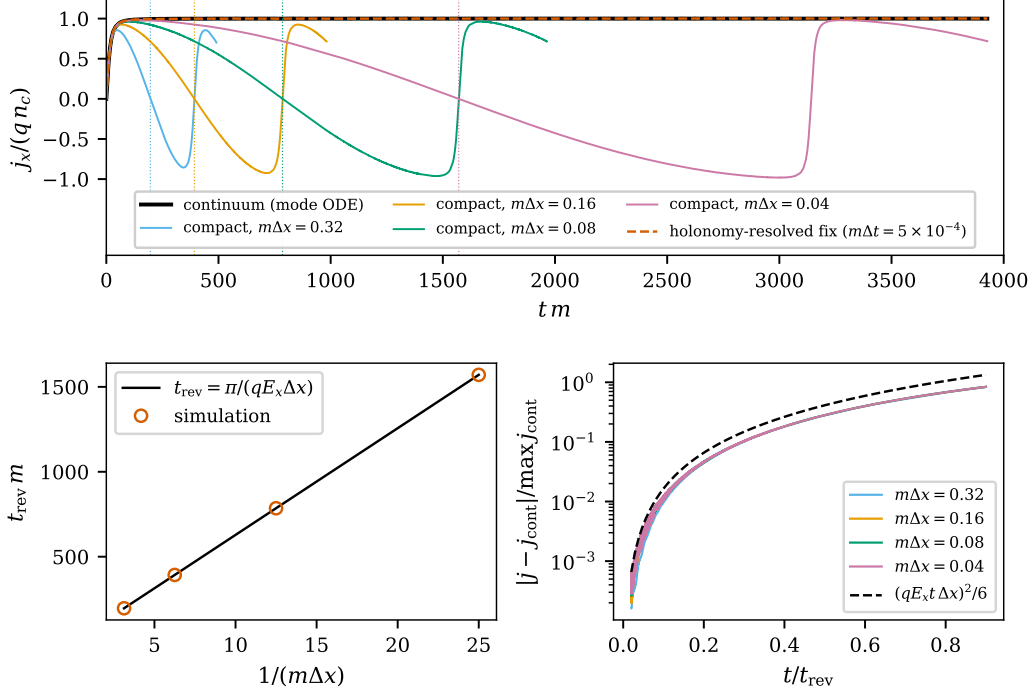


Figure 7: Sustained acceleration at constant force $F = qE_x = 0.05 m^2$. (a) Current of the dilute condensate: continuum reference (black, saturating at qn_c), compact-lattice runs at four resolutions (each reversing at its predicted $t_{\text{rev}} = \pi/(F\Delta x)$, dotted verticals), and the holonomy-resolved extension (dashed), which tracks the continuum across the entire window. (b) Measured reversal times versus $1/\Delta x$ against the parameter-free prediction. (c) Pre-reversal error growth at the four resolutions, collapsing onto the derived law $(qE_x t \Delta x)^2/6$.

maximum relative error $\{2.1 \times 10^{-2}, 5.9 \times 10^{-3}, 2.4 \times 10^{-3}\}$ at $m\Delta t = \{0.002, 0.001, 0.0005\}$, converging at second order in Δt as expected for the leapfrog action; mode-wise charge is conserved to 3.0×10^{-12} , and the exact large-gauge relabeling covariance $\bar{a}_1 \rightarrow \bar{a}_1 + 2\pi/qL$, $\hat{\phi}_k \rightarrow \hat{\phi}_{k-2\pi/L}$ holds to 3.7×10^{-14} with the dynamical mean Ampère law active. The multi-mode variant (Fig. 8) makes the same point for an extended momentum distribution: the compact scheme’s ensemble current departs from the per-mode continuum reference already at $t/t_{\text{rev}} \sim 0.3$ (band edges reach the zone boundary first) and reverses on schedule, while the extension stays within 8.1×10^{-3} of the reference over the full window (compact scheme: 1.2×10^{-1} over the same early window).

4.6 Summary of benchmarks

Table 1 collects the six studies: the two verifications against Shi et al. [2018] and the four benchmarks, with the conservation figures for one representative realization of each.

5 Analysis

One dichotomy organizes all six studies. Every phenomenon the scheme captured—linear branches, wakefields, threshold pair creation in a plasma, four-branch two-species spectra, superradiance—is *stimulated*: it develops from modes carrying macroscopic occupation in the initial data. The first boundary is the failure of *spontaneity* itself: vacuum pair creation has zero occupation anywhere to grow from, and the ε -scan turns the dichotomy into a measured law, $n_k = \varepsilon|\beta_k|^2$. This law is worth stating as the operational validity criterion of classical-statistical lattice QED: the scheme computes correctly those observables whose generating modes satisfy $n_k \gg \frac{1}{2}$, and undercounts any process fed by the half-quantum—by the factor $n_k/(n_k + \frac{1}{2})$ in linear response. The

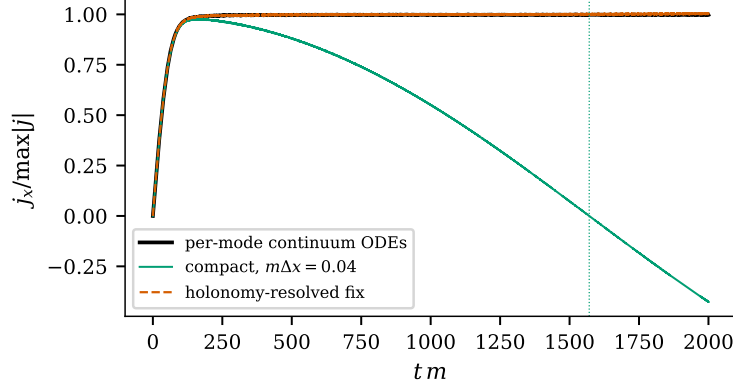


Figure 8: The horizon is not specific to condensates: a Gaussian band of matter modes ($\sigma_k = 2m$) accelerated by the same force. The compact scheme (solid) departs from the per-mode continuum reference (black) and reverses near $t_{\text{rev}}(k=0)$; the holonomy-resolved extension (dashed) follows the reference throughout.

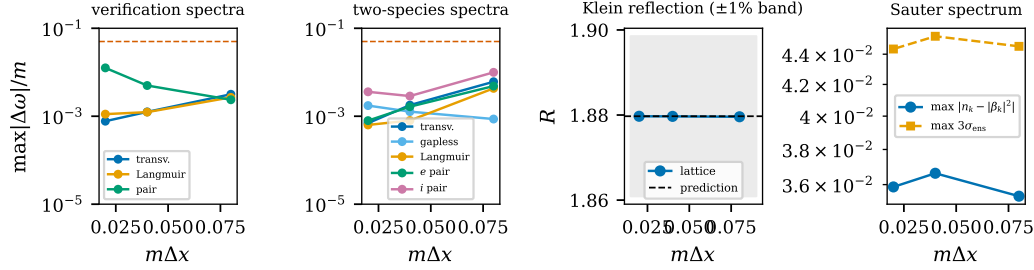


Figure 9: Convergence studies. (a) Single-species verification: peak deviations from the tree-level branches at three resolutions (tolerance dashed). (b) Two-species benchmark, same convention. (c) Klein reflection versus resolution against the mode-matching prediction. (d) Quantum-seeded Sauter spectrum: maximum deviation from the exact $|\beta_k|^2$ versus resolution against the ensemble bound. (Convergence for the acceleration benchmark appears in Fig. 7: reversal times across $8\times$ in Δx and the fix across $4\times$ in Δt .)

laser-plasma demonstration of Shi et al. [2018] lands on the safe side of the criterion—the pairs there are seeded by the plasma field itself, whose condensate occupations exceed unity by many orders of magnitude in the participating band—which our reproduction confirms; but the same simulation *silently omits* the spontaneous channel in its vacuum regions, and Fig. 6 quantifies exactly how much is missing and how to restore it. The reproduction itself illustrates the same sensitivity from the stimulated side: the above-threshold laser depletion is dominated by a backscatter-mediated channel that grows out of the microscopic fluctuation content of the initial ensemble, and two ensembles with identical macroscopic profiles but independently generated fluctuations (ours and the original) produce identical phenomenology at visibly different depletion rates. Observables fed by marginally occupied modes inherit the seeding sensitivity; the ε -law is the controlled statement of that fact, and quantum seeding is its principled resolution.

Conservation certifies the integrator, not the physics. Across the full battery, every structure-preservation diagnostic remained exact even where the physics failed: charge drifts between 10^{-15} and 10^{-13} , Gauss residuals from 10^{-13} (noise-seeded plasma runs) down to 10^{-22} (driven condensate), gauge covariance at the 10^{-14} – 10^{-13} level (exact twin; large-gauge relabeling), and the driven-run energy ledgers closing at their formal order. Structure preservation guarantees that the discrete system solves *its own* equations faithfully—which is what makes the failures attributable to the discrete *theory* rather than to its integration, and is precisely why a variational scheme is the right

Table 1: Summary of verification runs and benchmarks. Predictions and tolerances are those pre-stated in Section 2; $|\Delta Q/Q|$ and $|\Delta \mathcal{U}/\mathcal{U}|$ are the relative drifts of total charge and total discrete energy over one representative realization (*for the externally driven benchmarks—the Sauter pulse, the Klein step, and the sustained acceleration—the energy entry is the work–energy ledger residual, since the driving background makes the temporal-gauge energy genuinely non-conserved; for the Sauter pulse it is normalized by the peak external work, the matter energy being dominated by the inert seeded-vacuum background; †flagged entries are the quantum-seeded extension, $\varepsilon = 1$, quenched sector).

Test	Category	Regime probed	Prediction	Tolerance / failure criterion	Outcome	$ \Delta Q/Q $	$ \Delta \mathcal{U}/\mathcal{U} $
Paper Sec. III.1	verif.	unmagnetized 1D dispersion	$\omega^2 = \omega_p^2 + k^2$; $1 + \chi = 0$	$ \Delta \omega \leq 0.05m$	all branches within $0.005m$	4.9×10^{-15}	1.8×10^{-11}
Paper Sec. III.2	verif.	1D laser–plasma, $a \approx 1, 16$	wakefield + Schwinger threshold; GO transit	transit $\pm 5\%$	GO transit dev. 1.1%; phenomenology incl. pair burst reproduced (depletion seed-limited, see text)	3.9×10^{-14}	3.5×10^{-4}
Pair-ion two-species	scope	multi-species collective modes	4-branch $1 + \chi_e + \chi_i = 0$	$ \Delta \omega \leq 0.05m$	all resolvable branches within $0.002m$	4.7×10^{-15}	1.4×10^{-11}
Sauter-barrier superradiance	scope	supercritical static field, Klein zone	$R_{\text{pred}} = 1.8797$	$ \Delta R \leq 1\%$	$R = 1.8797$ (dev. 0.002%); gauge twin exact	9.4×10^{-14}	$7.9 \times 10^{-5*}$
Vacuum Sauter pulse	boundary	spontaneous pair creation, $E_0 = 1.5E_S$	QFT: $n_k = \beta_k ^2$; classical: $n_k = \varepsilon \beta_k ^2$	fail if ε -linear, no floor; fix: Eq. (10)	failure confirmed ($\varepsilon=0$: yield 0); fix verified (spectrum within bound; yield dev. 0.7%)	$1.8 \times 10^{-14\dagger}$	$4.9 \times 10^{-3*\dagger}$
Sustained acceleration	boundary	secular drive, $\gamma \gg 1$	continuum: $j \rightarrow qn_c$ monotone; lattice: reversal at t_{rev} of Eq. (12)	deep-reversal criterion; fix: err $\leq 1\%$	reversal at all Δx (ratio = 1.000); fix tracks to $\gamma \approx 196$ at 0.24%	1.7×10^{-13}	$4.4 \times 10^{-2*}$

laboratory for this kind of boundary-mapping. The corollary for practitioners is sharp: conservation traces cannot detect either boundary; only confrontation with analytic strong-field references can.

The two extensions insert the two missing scales. The quantum-seeded ensemble inserts $\hbar$ (the commutator floor $\varepsilon = 1$); the holonomy-resolved sector inserts noncompactness (the unbounded kinetic momentum of continuum torus QED). Neither touches the variational structure: both are discrete actions, both conserve charge exactly, and both were verified against exact references at multiple resolutions. Their costs are complementary and quantifiable. Seeding loads every mode with energy $\omega_k/2$, a lattice vacuum energy density $\frac{1}{L} \sum_k \frac{\omega_k}{2} \rightarrow 2/(\pi \Delta x^2)$ in the massless limit, growing quadratically under refinement; in the linear (quenched) regime it is inert and exactly subtractable, but with backreaction it feeds secular classical (Rayleigh–Jeans) heating, which is the known frontier of classical-statistical methods [Aarts and Smit, 1999, Berges et al., 2014, Mou et al., 2019]—our self-consistent run measures the induced mean field at $\langle \delta E_x \rangle / E_0 = 4.1 \times 10^{-1}$ by the end of the pulse, already a leading-order backreaction at this vacuum load, so quantitative self-consistent evolutions require the subtraction schemes explored in that literature. The holonomy extension trades site-locality of one global mode for unbounded momenta and shifts the burden to the time step, $\Delta t \lesssim 1/\omega_{\text{max}}(T)$: cost *linear* in the target momentum, replacing impossibility ($p_{\text{max}} \Delta x < \pi$ at fixed Δx) with expense. The residual time-compactness is visible in our own data: a logged stability scan of the extension finds the $m\Delta t = 0.02$ run diverging at $t \approx 2002/m$ —precisely where $\omega_{\text{max}}(t) \Delta t$ crosses 2.00, the leapfrog bound—while $m\Delta t = 0.01$ survives formally ($\omega_{\text{max}}(T) \Delta t = 1.96$) yet is dominated by order-unity phase error at the stability edge (maximum current error 4.3×10^0), and $m\Delta t = 0.005$ still errs at 1.5×10^{-1} ; quantitative accuracy requires the production steps $m\Delta t \leq 0.002$ of Section 4.5. The horizon is therefore not destroyed but displaced onto the time lattice, where—unlike in space—refinement is cheap in one dimension.

Where the original demonstrations sit inside the map. The compactness horizon quantifies which accelerator problems the original scheme can address at a given resolution: by the collapse law of Fig. 7c the current error is $(p\Delta x)^2/6$, so momenta are faithful at the 2% level only up to $p\Delta x \lesssim 0.35$ ($\gamma \lesssim 9$ at the published resolution $m\Delta x = 0.04$); an 18% tolerance stretches this to $p\Delta x = \pi/3$ ($\gamma \lesssim 26$), and the hard horizon—current reversal—sits at $p\Delta x = \pi$ ($\gamma \approx 79$). Our reproduction of the $a \simeq 16$ run shows transverse spectral content extending to a few m (Fig. 3) and matter energies per particle far below the band edge—safely inside—but a wakefield stage aiming at $\gamma \sim 10^2$ – 10^3 , or any sustained-field problem (capacitor discharge, magnetar gap electrodynamics), crosses the horizon at any affordable Δx ; the holonomy-resolved sector, or a hybrid in which it carries the mean accelerating field while compact links carry fluctuations, is then not an optimization but a requirement. These boundaries matter for the growing program of strong-field and quantum-plasma simulation [Shi, 2021, Gonoskov et al., 2022, Fedotov et al., 2023, Brodin and Zamanian, 2022]. Conversely, the ε -law warns that quantitative pair yields in the *wings* of laser pulses (where plasma occupation dies off) are systematically undercounted by the unseeded scheme even though in-plasma yields are right; a seeded ensemble restores the wings at the cost of the vacuum-energy load above.

Unexpected observations. Three findings were not planned but emerged from the data. First, the cold two-species system supports a gapless branch of purely quantum origin (the cold classical two-fluid limit has none); balancing the recoil terms of the two susceptibilities yields the closed-form long-wavelength law $\omega \rightarrow k^2/(2\sqrt{m_e m_i})$ [Eq. (5)], a single-particle recoil frequency for the geometric-mean mass from which the plasma frequencies cancel entirely, and the measured spectra resolve the crossover from this recoil asymptote onto the full quartic; the branch should appear in any future two-species application of the scheme. Second, the Wilson-line bound $qA\Delta < 2\pi$ quoted by Shi et al. [2018] is saturated at *half* its value: physics fails at $p\Delta x = \pi$ (group-velocity turnover), not at 2π (full phase wrap)—the practically relevant constraint is twice as strict. Third, the energy defect of the coupled scheme is not merely bounded by $\mathcal{O}(q\Delta^2)$: in the driven benchmark its prefactor is set by $(\Omega_{\max}\Delta t)^2$ with $\Omega_{\max}$ the *highest occupied* frequency, which our Δt -halving isolates cleanly; this makes the paper’s own 0.2% figure at $a \simeq 16$ quantitatively predictable from the spectral content of the run.

Limitations. This study is one-dimensional; transverse compactness (the same Wilson-line periodicity in Δy , relevant at $a \gtrsim 2\pi/(qm\Delta y) \sim 150$ for the published resolutions) and magnetized geometries remain untested, though nothing in the analysis is specific to 1D. The matter field is scalar: spin-statistics effects—Pauli blocking of produced pairs, the factor-of-two vacuum-energy difference—require the spinor extensions of [Chen et al., 2021, Kasper et al., 2014] and are quantitatively assessed for pair creation in [Al-Naseri and Brodin, 2023]. The classical-statistical truncation itself omits loop effects beyond the seeded tree level, and our seeded runs were verified in the linear/quenched regime plus one self-consistent configuration; the long-time UV dynamics of seeded ensembles was bounded but not mapped. The holonomy-resolved extension is implemented for spatially uniform accelerating fields (arbitrary matter spectra); its coupling to inhomogeneous dynamical gauge fluctuations is formulated but unimplemented, and we expect the split to inherit the standard difficulties of separating scales in gauge theories. Finally, all ensembles here are modest (≤ 2000); we quote ensemble errors throughout, and none of the conclusions strains them.

6 Conclusion

We mapped, with pre-stated quantitative criteria, what the variational real-time lattice formulation of scalar QED does and does not compute correctly for strong-field plasma physics. On the positive side the verified scope now includes fully dynamical multi-species plasmas—with a four-branch relativistic-quantum spectrum containing a gapless quantum-recoil branch, $\omega \rightarrow k^2/(2\sqrt{m_e m_i})$ at long wavelength—and superradiant Klein scattering with exact charge book-keeping, both matching tree-level theory at the percent level or better. On the negative side we isolated two structural boundaries, invisible to every conservation diagnostic the scheme so carefully preserves: classical-statistical evolution cannot originate spontaneous processes (measured law $n_k = \varepsilon|\beta_k|^2$, no classical floor), and compact Wilson-line kinematics cannot represent secular acceleration (measured horizon $t_{\text{rev}} = \pi/(qE\Delta x)$ at every resolution). Both boundaries were then *crossed* by extensions that stay inside the variational, exactly-charge-conserving family: Wigner half-quantum seeding with symplectic lattice normalization, which reproduces exact Sauter–Schwinger spectra mode by mode;

and a holonomy-resolved spectral matter sector, which restores uniform-in-time convergence to $\gamma \approx 200$ at 0.24%. The practical upshot for strong-field simulation is a validity map with tested exits: occupation decides whether a process exists in the simulation, compactness decides how far it can be accelerated, and both deficits are repairable without sacrificing the structure that makes the method trustworthy in the first place. Natural next steps are the spinor analogue of the seeded ensemble, the hybrid holonomy-plus-links scheme for inhomogeneous wakefields, and two-dimensional magnetized benchmarks, for which the present four studies supply the reference points; the same benchmarks, being exactly solvable, are natural acceptance tests for the emerging quantum-hardware implementations of lattice gauge dynamics [Feynman, 1986, Lloyd, 1996, Wiese, 2013, Martinez et al., 2016, Sundar et al., 2025, Shi et al., 2024].

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